

a.

	total	
1	33	
	(No column name)	
1	3789.3	
	(No column name)	
1	164.752173	
	(No column name)	
1	62.9	
	(No column name)	
1	73.2	
	(No column name)	
1	JOHN	
2	MARY	
	(No column name)	
1	smith	
2	johnson	
	(No column name)	
1	4	
2	4	
	(No column name)	
1	164.750000	
	(No column name)	
1	2026-04-30 10:12:25.343	
	(No column name)	
1	John Smith	
2	Mary Johnson	
	(No column name)	
1	2	

b.

6. Like in where clause

```
1 SELECT * FROM Patient Where FName LIKE 'J%';
2
```

0 % ✓ No issues found

Results  Messages

	PatientID	FName	LName	MName	DOB	Gender	Height	Weight	
	1	John	Smith	A	1985-03-12	M	70.5	180.2	
	9	James	Harris	K	1984-04-03	M	68.9	170.3	
	15	Joseph	Clark	B	1986-07-08	M	69.8	182.1	

a.

7. Row Count

```
1 Set RowCount 4;
2 Select * From Patient;
3 Set RowCount 0;
4 Select * From Patient;
5 Select Top 4 * From Patient;
6
```

0 % ✓ No issues found

Results Messages

	PatientID	FName	LName	MName	DOB	Gender	Height	Weight
	1	John	Smith	A	1985-03-12	M	70.5	180.2
	2	Mary	Johnson	L	1990-07-25	F	64.2	135.6
	4	Patricia	Taylor	M	1982-01-18	F	66.8	150.4
	6	Linda	Thomas	R	1988-09-14	F	63.5	140.2
	PatientID	FName	LName	MName	DOB	Gender	Height	Weight
5	18	Betty	Lee	C	1989-09-01	F	64.0	136.3
7	19	Charles	Walker	D	1983-12-14	M	71.0	188.8
3	20	Sandra	Hall	NULL	1996-04-26	F	65.4	149.2
	PatientID	FName	LName	MName	DOB	Gender	Height	Weight
	1	John	Smith	A	1985-03-12	M	70.5	180.2
	2	Mary	Johnson	L	1990-07-25	F	64.2	135.6
	4	Patricia	Taylor	M	1982-01-18	F	66.8	150.4
	6	Linda	Thomas	R	1988-09-14	F	63.5	140.2

a.

8. Ailment Procedure

```

1  Create Procedure InsertAilment
2      @AilmentID INT,
3      @Ailment nvarchar(50),
4      @AilDesc nvarchar(MAX)
5  AS
6  Begin
7      Insert Into Ailment (AilmentID, Ailment, AilDesc)
8      Values (@AilmentID, @Ailment, @AilDesc);
9  END;
10 EXEC InsertAilment 10, 'Cold', 'Common cold';

```

a.

9. Middle Name Update

```

1  Create Procedure UpdatePatientMiddleName
2      @PatientID INT,
3      @MName nvarchar(50)
4  AS
5  Begin
6      Update Patient
7      Set MName = @MName
8      Where PatientID = @PatientID;
9  END;
10 EXEC UpdatePatientMiddleName 1, 'Michael';

```

a.

10. Create Table and Add Data

```

1  Create Table SQL1Table (
2      SampleID INT,
3      SampleName nvarchar(50),
4      SampleValue INT
5  );
6  Truncate Table SQL1Table;
7  Insert Into SQL1Table Values
8  (1, 'A', 10), (2, 'B', 20), (3, 'C', 30), (4, 'D', 40), (5, 'E', 50),
9  (6, 'F', 60), (7, 'G', 70), (8, 'H', 80), (9, 'I', 90), (10, 'J', 100);
10 Select * From SQL1Table

```

0% 1 0 | ↑ ↓

Results Messages

SampleID	SampleName	SampleValue
1	A	10
2	B	20
3	C	30
4	D	40
5	E	50
6	F	60
7	G	70
8	H	80
9	I	90
10	J	100

a.

11. Insert Ailments for Patients

```

1  INSERT INTO PatientAilment (PatientID, AilmentID) VALUES (1,3), (1,2);
2  INSERT INTO PatientAilment (PatientID, AilmentID) VALUES (2,1), (2,3);

```

a.

12. Group By

```

1  ✓ Select P.PatientID, P.FName, COUNT(*) AS AilmentCount
2  From Patient P
3  Join PatientAilment PA ON P.PatientID = PA.PatientID
4  Group By P.PatientID, P.FName;

```

100 % ✓ No issues found

Results Messages

	PatientID	FName	AilmentCount
1	1	John	4
2	2	Mary	4
3	4	Patricia	1
4	6	Linda	1
5	7	William	2
6	8	Barbara	1
7	9	James	2

a.

13. Substring

```

1  Select FName, Substring(FName,1,3)
2  From Patient;

```

100 % ✓ No issues found

Results Messages

	FName	(No column name)
1	John	Joh
2	Mary	Mar
3	Patricia	Pat
4	Linda	Lin
5	William	Wil
6	Barbara	Bar

a.

14. Distinct and Order

```

1  -- Ascending order
2  ✓ Select Distinct LName
3  From Patient
4  Order By LName ASC;
5
6  -- Descending order
7  ✓ Select Distinct LName
8  From Patient
9  Order By LName DESC;

```

100 % ✓ No issues found

Results Messages

	LName
1	Allen
2	Clark
3	Garcia
4	Hall
5	Harris

	LName
1	Young
2	Wright
3	White
4	Walker
5	Thompson

a.

15. @@version

1	Select @@version AS SQLServerVersion;
2	
100 % No issues found	
Results Messages	
SQLServerVersion	
1	Microsoft SQL Server 2025 (RTM-GDR) (KB5084814)...

a.

16. Ailment Count

1	Select Count(*) AS TotalAilments
2	From Ailment;
100 % No issues found	
Results Messages	
TotalAilments	
1	25

a.

17. Most Ailments

1	Select Top 1 P.PatientID, P.FName, P.LName, Count(*) AS AilmentCount
2	From Patient P, PatientAilment PA
3	Where P.PatientID = PA.PatientID
4	Group By P.PatientID, P.FName, P.LName
5	Order By AilmentCount DESC;
100 % No issues found	
Results Messages	
PatientID FName LName AilmentCount	
1	1 John Smith 4

a.

18. Longest Last Name

1	Select Top 1 ProviderID, FName, LName,
2	Len(LName) AS LastNameLength
3	From Provider
4	Order By Len(LName) DESC;
100 % No issues found	
Results Messages	
ProviderID FName LName LastNameLength	
1	6 Laura Hernandez 9

a.

19. Insert Patient with Multiple Ailments

1	-- Insert Patient
2	Insert Into Patient (PatientID, FName, LName, MName, DOB, Gender, Height, Weight)
3	Values (101, 'Alice', 'Walker', NULL, '1990-06-15', 'F', 65.5, 140.0);
4	
5	-- Insert Multiple Ailments for that Patient
6	Insert Into PatientAilment (PatientID, AilmentID, StartDate, EndDate, ProviderID)
7	Values
8	(101, 1, '2025-01-01', NULL, 1),
9	(101, 2, '2025-02-01', NULL, 1),
10	(101, 3, '2025-03-01', NULL, 2);
11	
12	-- Display Patient with Ailments (Implied Join)
13	Select P.PatientID, P.FName, P.LName, A.Ailment, PA.StartDate, PA.EndDate
14	From Patient P, PatientAilment PA, Ailment A
15	Where P.PatientID = PA.PatientID
16	and PA.AilmentID = A.AilmentID
17	and P.PatientID = 101;
100 % No issues found	
Results Messages	
PatientID FName LName Ailment StartDate EndDate	
1	101 Alice Walker Hypertension 2025-01-01 NULL
2	101 Alice Walker Diabetes Mellitus 2025-02-01 NULL
3	101 Alice Walker Asthma 2025-03-01 NULL

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